

Abhijit Rajkumar

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PROJECTS

GitHawk | *Next.js, Better Auth, Pinecone, Gemini AI, Inngest, Polar, PostgreSQL* | [GitHub](#) | [Live Demo](#)

- Built a full-stack AI-powered GitHub Pull Request reviewer — an open-source alternative to CodeRabbit — using Next.js, Better Auth (GitHub OAuth), and PostgreSQL via Prisma.
- Implemented a **RAG pipeline** using Pinecone vector embeddings to index entire repository codebases and retrieve semantically relevant context for every PR review.
- Engineered an Inngest background job that fetches the PR diff, runs vector search, generates a structured review with Gemini 2.5 Flash (walkthrough, Mermaid diagram, bug analysis, suggestions), and posts the result as an inline GitHub PR comment — with **concurrency control and automatic retries**.
- Integrated Polar for subscription management and built a separate index-repo Inngest function to embed all repo files into Pinecone on first repository connection.

v0-clone | *Next.js, Inngest Agent Kit, GPT-4.1, E2B Sandboxes, Clerk, PostgreSQL* | [GitHub](#) | [Live Demo](#)

- Built a full-stack AI UI generation platform inspired by v0.dev — users describe a feature in plain English and receive a live, interactive Next.js preview running inside a real cloud sandbox.
- Architected an **Inngest Agent Kit coding agent** (GPT-4.1-mini) with terminal, createOrUpdateFiles, and readFiles tools that autonomously scaffolds, writes, installs dependencies, and debugs code inside E2B cloud sandboxes.
- Implemented sandbox reuse logic that reconnects to an existing E2B sandbox across conversation turns, preserving installed node modules and prior file state to eliminate redundant reinstalls.
- Persisted every generation as a Fragment (file snapshot + live URL) linked to a project message thread, enabling full multi-turn iterative refinement with complete history via Prisma and PostgreSQL.

Patho-Predict | *Next.js, Modal, FastAPI, Biopython, UCSC API, ClinVar, Pinecone* | [GitHub](#) | [Live Demo](#)

- Developed a full-stack genomic variant pathogenicity prediction platform — input a chromosomal position and alternative nucleotide, receive an AI-powered clinical classification with evidence.
- Built a **Modal serverless FastAPI endpoint** that fetches the reference base from the UCSC Genome Browser API, translates the codon change into an amino acid substitution with Biopython, and scores the variant across three signals: CADD-like transition/transversion penalty, biochemical AA severity, and GC-context conservation.
- Designed a weighted scoring model ($\text{risk} = 0.45 \times \text{CADD} + 0.35 \times \text{ESM-proxy} + 0.20 \times \text{Conservation}$) with thresholded output into Likely Pathogenic, Uncertain Significance, or Likely Benign.
- Integrated ClinVar variant overlays, gene sequence visualization, and 7 Next.js API routes consuming UCSC and NCBI APIs — supporting any UCSC genome assembly (hg38, hg19, mm39, etc.).

WORK EXPERIENCE

AI / ML Intern

Sep 2025 – Apr 2026

Euron (Engagesphere Technology Pvt Ltd) | [Certificate](#)

Remote

- Built **MediRAG (RAGnosis)**, a full-stack AI-powered medical document analysis system using **Retrieval-Augmented Generation (RAG)**.
- Designed end-to-end pipeline: document ingestion → embeddings (**LlamaIndex, HuggingFace**) → vector search (**Pinecone/FAISS**) → LLM response generation.
- Developed scalable architecture using **FastAPI backend, Next.js frontend**, and **Docker Compose** for deployment.
- Implemented semantic search and source-grounded responses for accurate medical Q&A using LLMs.

Apprentice

Jan 2022 – Nov 2023

Pwskills

Bengaluru, Karnataka

- Led development of **machine learning solutions** for EDA (Exploratory Data Analysis) products, focusing on innovative and efficient algorithms.
- Collaborated with cross-functional teams including data scientists and software developers to integrate ML capabilities into EDA tools.
- Enhanced **problem-solving skills** by actively engaging in troubleshooting and resolving complex engineering challenges.

AI / ML Intern

Jun 2021 – Aug 2021

Elite Techno Group

Jaipur, Rajasthan

- Developed a Breast Cancer Prediction model achieving a classification accuracy of **99.45%**.
- Engineered a **modular ML pipeline** with distinct components for data ingestion, transformation, model training, and prediction.
- Gained hands-on experience in end-to-end machine learning workflow development and optimization.

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, C, C++, R
- **Full-Stack:** Next.js, React, FastAPI, Node.js, Express.js, Tailwind CSS, REST APIs, Streamlit
- **AI & LLMs:** LangChain, OpenAI GPT, Google Gemini, Inngest Agent Kit, RAG, Pinecone, Qdrant, HuggingFace
- **ML:** Supervised & Unsupervised Learning, Neural Networks, CNNs, Random Forest, XGBoost, Time Series
- **Databases:** PostgreSQL, MongoDB, MySQL, SQLite, Pinecone, Qdrant, Prisma
- **Big Data:** Hadoop, PySpark, Apache Kafka, Hive, HDFS, AWS EMR
- **Cloud & DevOps:** AWS (EC2, S3, Lambda, RDS), GCP (BigQuery), Docker, Kubernetes, GitHub Actions, Nginx
- **Tools:** E2B Sandboxes, Modal, Inngest, Better Auth, Clerk, Polar, Biopython, BeautifulSoup, Postman

EDUCATION

Gurucharan College

Silchar, Assam

Bachelor of Science in Mathematics

Jun 2013 – Jun 2017

- **Core Subjects:** Linear Algebra, Real Analysis, Number Theory, Complex Analysis, Mathematical Modeling
- Applied mathematical concepts to real-world problems in traffic control and rainfall prediction.

SCHOLASTIC ACHIEVEMENTS

- Completed an MSc project on **"Traffic Control Optimization using Linear Algebra,"** demonstrating expertise in mathematical modeling. Actively working towards publication.
- Completed an MSc project on **rainfall prediction using Time Series analysis**, applying advanced statistical techniques to real-world challenges. Pursuing academic publication of findings.

CERTIFICATIONS

Big Data Engineering Intern

Sep 2025 – Apr 2026

Euron (Engagesphere Technology Pvt Ltd) | [View Certificate](#)

Remote

- Hadoop, Spark (RDDs, DataFrames, Catalyst), Kafka, Spark Streaming. Orchestration with **Airflow and Azure Data Factory**.
- Cloud: **Azure** (Data Lake, Databricks), **AWS** (S3, EMR, Glue, Athena). Projects: Fraud Detection, Sentiment Analysis.

Data Science Masters Certification

Nov 2023

PW Skills | [View Certificate](#)

Remote

- End-to-end ML workflows; Deep Learning with **TensorFlow, Keras, PyTorch**; PCA, t-SNE, LDA for feature engineering.
- Domain projects in healthcare, finance, and retail using **Random Forest, XGBoost, SVM**, and Logistic Regression.

Full Stack Developer Cohort (0 to 100)

July 2024

Instructor: Harkirat Singh | [View Certificate](#)

Remote

- MERN stack, System Design, DevOps (**Docker, Kubernetes, CI/CD, Nginx, Prometheus, Grafana, AWS**).
- Advanced: Kafka, WebSockets, gRPC, Rate Limiting. Projects: Paytm Clone, Zerodha Clone, Zapier Clone.